



- 1 Tick (✓) **one** box to complete the description of abstraction.

Abstraction is part of the analysis stage of the development life cycle. Abstraction means

- A** testing the solution to the problem.
- B** writing a program using program code.
- C** defining the sub-systems that make up the problem.
- D** removing details from the problem that are not relevant.

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[1]

- 2 A fence measuring 2 metres in height and 20 metres in length needs painting. The area of the fence needs to be calculated so that enough paint for the job is purchased.

After the task is decomposed, there are three component parts: input, process and output.

**Three** component parts and **four** example descriptions are shown.

Draw **one** line from each component part to the most appropriate relevant description.

**Not** all descriptions will be used.

| Component part | Description                                             |
|----------------|---------------------------------------------------------|
| process        | the height of the fence<br>the length of the fence      |
| output         | The amount of paint required is<br>40 square metres.    |
| input          | the colour of the paint                                 |
|                | The area of fence is calculated as<br>40 square metres. |

[3]

- 3 A range check is used to validate that the input temperature is an integer between –100 and 0, inclusive.

Suggest suitable normal, abnormal, extreme and boundary test data to test if this validation check is working correctly.

Normal test data .....

Abnormal test data .....

Extreme test data .....

Boundary test data .....

[4]

